

Problem Set 5 Solutions

QTM 220

Due 3 November at 2:00pm ET on Canvas. Questions about submission requirements should be directed to your TA.

Gerber, Green, and Larimer Michigan Voter Experiment

Prior to the August 2006 primary election in Michigan, political scientists Gerber, Green, and Larimer conducted a large field experiment in which different mailers were sent to registered voters across Michigan. Because whether or not someone votes (though not who they vote for) is public record, the researchers then accessed the public voting information to measure their outcome— whether an individual voted in the August 2006 primary. The different treatments are as follows:

- Civic Duty: do your civic duty and vote!
- Hawthorne: you are being studied + do your civic duty and vote!
- Self: who votes is public information, here is your recent voting record + do your civic duty and vote!
- Neighbors: what if your neighbors knew whether you voted? here is your recent voting record + your neighbors' recent voting record + do your civic duty and vote!
- Control: no mailer

Households were randomized to receive a mailer (or not) and which type, but voting records are by individual.

The attached data from their experiment has the following variables:

- hh_id : the household id number
- hh_size : number of adults in household
- sex : individual's sex according to voter record

- `yob`: individual's year of birth
 - `g2000` : whether or not the individual voted in the general election in 2000
 - `g2002` : whether or not the individual voted in the general election in 2002
 - `g2004` : whether or not the individual voted in the general election in 2004
 - `p2000` : whether or not the individual voted in the primary election in 2000
 - `p2002` : whether or not the individual voted in the primary election in 2002
 - `p2004` : whether or not the individual voted in the primary election in 2004
 - `treatment` : the mailer that the household received (or Control for no mailer)
 - `voted` : whether the individual voted in the 2006 primary (outcome)
- (a) Create a balance table showing the similarity of individuals assigned to treated and control across sex, age, household size, and turnout in previous elections.
 - (b) Create a balance table showing the similarity of individuals assigned to each level of treatment and control across sex, age, household size, and turnout in previous elections. Be sure to note how many individuals are in each level of treatment.
 - (c) Calculate the subsample proportion of 2006 primary turnout within each level of treatment (and control). Plot?
 - (d) Let's define the following treatment effects of interest: $\hat{\tau}(civic) = \hat{\mu}(civic) - \hat{\mu}(control)$, $\hat{\tau}(hawthorne) = \hat{\mu}(hawthorne) - \hat{\mu}(civic)$, $\hat{\tau}(self) = \hat{\mu}(self) - \hat{\mu}(hawthorne)$, $\hat{\tau}(neighbor) = \hat{\mu}(neighbor) - \hat{\mu}(self)$. Calculate each of these and interpret each in plain language.
 - (e) Create a bootstrap estimated sampling distribution for each of your τ s and a 95% confidence interval for each.
 - (f) Fit a line? Why is it bad?
 - (g) Bootstrap estimated sampling dist for line? what happens to precision?
 - (h) BONUS?: Why might we be concerned about using individual turnout as the outcome for this experiment?

Women's Health Initiative

describe data, variable names

- (a) Balance table on age
- (b) Overall estimated treatment effect (difference in proportions) of CEE+MPA on breast cancer
- (c) Bootstrap estimated sampling distribution, 95% confidence interval
- (d) Estimate treatment effect within each age group
- (e) bootstrap estimated sampling distribution for within-group treatment effects, 95% confidence intervals